



Waipapa
Taumata Rau
**University
of Auckland**



ENGGEN 403 Team Project Brief: speed limits

[24-28 August 2026]

Speed limits

Assignment Objectives

1. To critically assess evidence and draw conclusions to inform your recommendations
2. To practice business case analysis
3. To discover which online collaboration tools work best for your team

Project Background

With a few exceptions (Gossling et al, 2023) most countries in the world have speed limits. In Aotearoa New Zealand, speed limits are a contentious issue that polarise public and political opinion. In this project you will attempt to separate the evidence on the impacts of setting different speed limits from the values-driven debates that surround them.

The arguments to increase speed limits tend to centre on increasing productivity (see, for example: <https://www.gov.uk/government/news/speed-limit-change-helps-increase-productivity>). The degree to which enabling faster driving does save time and increase productivity has been questioned (Kingham, 2022, 2024, and Welch, 2025) and is likely to have different effects in different settings (e.g. on the open road, compared to central cities).

The arguments to decrease speed limits tend to centre on safety to both drivers and passengers and other road users, since it is well established that road accidents cause more harm at higher speeds (Kingham, 2022; Yannis and Michelaraki, 2024, 2025; Quddus et al, 2025).

Environmental arguments also come into play, with evidence suggesting that optimising speed limits can reduce fuel consumption and emissions (Gossling et al, 2023). There are also costs and benefits associated with pollution and noise, and the subsequent impact on health (Kingham, 2022).

Quantifying the impact of changing speed limits is challenging and requires detailed cost benefit analysis (Gossling et al, 2023). Assessing the relative importance of productivity benefits, safety, economic and environmental issues requires value judgements which inevitably influence how the cost benefit analysis is undertaken.

Government context:

In New Zealand, the Government's transport policy and what it wants us to achieve were until recently set out by the Ministry of Transport in the Transport Outcomes Framework and the Government Policy Statement on Transport (<https://www.transport.govt.nz/area-of-interest/strategy-and-direction/transport-outcomes-framework>). More broadly, the Ministry of Transport sought to ensure New Zealanders' wellbeing and quality of life by improving safety, health, economic prosperity, environmental sustainability, and inclusive access. These functions have now been absorbed into a larger Ministry of Cities, Environment, Regions and Transport, with responsibility to develop legislation, oversee transport Crown entities, manage transport funding, and provide strategic direction for all transport modes to ensure a resilient, secure, fair, and efficient national transport system.

The Ministry has a national modelling team to provide guidance to the Ministry and the Minister on the impacts of policy changes (interventions) where a) the interventions are wide, e.g. tolling, new roads,

population change, broken roads, behaviour change etc and b) impacts of interest are also wide, e.g. travel duration, distance, emissions, network volumes, fairness, safety, satisfaction by/on whom where whom is defined by a combination of demographics and activities (purposes, locations).¹ The model includes the capacity to interrogate the impact of speed limits on a variety of economic, environmental, and social outcomes.

Waka Kotahi ([Our purpose | NZ Transport Agency Waka Kotahi](#)) is New Zealand's Transport Agency and has a role to manage land transport in New Zealand. To do so it must follow the laws and government policies about land transport and put the government's transport policies into practice.

Local government organisations also have a say in setting speed limits. For example, Auckland Transport has gathered an evidence base on safe speed limits, which is summarised here: [Safe speeds - the reasons](#).

Recent changes to speed limits

Ahead of the 2020 decreases in speed limits, Waka Kotahi tasked local councils with developing speed management plans to reduce transport-related deaths as part of a 'Road to Zero' project which aimed to reduce transport-related deaths. In many places, this resulted in a decrease of the speed limit in urban centres to 30 km/h (see for example: <https://at.govt.nz/about-us/transport-plans-strategies/vision-zero-for-the-greater-good/safe-speeds-programme/speed-limit-changes-around-auckland>) in line with international evidence that this shift saves lives (Yannis and Michelaraki, 2024,2025; Quddus et al 2025). These changes were reversed in 2024 (Kingham, 2024, and see: <https://www.nzta.govt.nz/safety/driving-safely/speed/state-highway-speed-management/speed-reversals-and-consultation-transitional-changes-in-2024-25>).

Problem Statement

Your team has been approached by a road safety advocacy group who are lobbying the Minister of Transport and his officials in the Ministry. A video briefing is available in the lecture slides posted on CANVAS.

You are tasked with examining **part a and one of b, c or d** and providing independent advice to the advocacy group who will likely provide your report to the Minister:

- a) Economic versus social versus environmental impacts of speed limit increases.
- b) Speed limits in Europe (and the UK) – lessons learnt for our New Zealand context.
- c) Technological solutions to traffic management in Aotearoa New Zealand – how might these be integrated to provide a holistic control solution that does not rely on speed limits as the only tool?
- d) The interplay between implementation in urban, regional and rural areas

As you explore your chosen issues, your team should identify two specific problems that Aotearoa New Zealand *must* address to successfully manage traffic.

¹ From a talk by at Capital City Complex Systems conference, 2026, Jade MacKay

Following the exploration of the problem space and development of two problem statements, your team should identify the Critical Success Factors for addressing the proposed challenge. Using the methodologies taught in class, the team should develop a list of options Aotearoa New Zealand could consider for addressing the two problems. These options should then be evaluated, and a short list of two packages returned to the client.

Assessment Outline

The 'team-focused' objectives of this assessment include figuring out what works for your team. Simultaneously, you are tasked with analysing a complex problem with the intent of developing your own proposal. You must identify and analyse the problem, the stakeholders, develop some options and make a recommendation of three shortlisted options (do minimum, do medium, do maximum), alongside do nothing, for the government to investigate further.

While some research is required to explore the problem space, helpful links and references have been provided (see references and additional resources on Canvas) to minimise the need for your whole team to conduct intensive research. Remember, this is not primarily a research paper, you will not all have sufficient time to fully review and digest all reference materials when creating this summary. Consider strategies such as delegating deep research to a small number of team members, and having them share key facts that help to underpin, inform, or support the wider team's understanding. This is an opportunity to analyse a problem and recommend potential improvements. Be thoughtful and creative, keeping in mind your limited time.

There is no right answer for the task at hand. Focus on understanding the problems; analysing available information; synthesising opportunities for change and innovation; and conveying to the reader why your recommendation is reasonable. Remember, your team will need to weigh up the information and take positions on approaches that are uncertain and sometimes ambiguous to complete the project.

All material should be presented in your own words. Please cite all sources your team uses using the APA 7thed format.

Due Dates and Submission:

- Due August 28 at 5 pm
- Submit via Canvas as a PDF (one submission per team)
- Report file name: [Team Number]_Team_Project_Report.pdf
- Reports will be assessed via Turn-it-in.
- See Team project Deliverable Document for more information.

Workload and Student Support

The effort required to complete this report is estimated at 8 to 10 hours per student. The teaching team will be available from 10 am – 4 pm each day in the 'clinic' in 405-921. Ed Discussions will be operating as usual, and all questions will be answered as promptly as possible.

References

- Auckland Council 2020: <https://at.govt.nz/about-us/transport-plans-strategies/vision-zero-for-the-greater-good/safe-speeds-programme/speed-limit-changes-around-auckland>
- Auckland Transport: <https://at.govt.nz/about-us/transport-plans-strategies/vision-zero-for-the-greater-good/safe-speeds-programme/safe-speeds-the-reasons>
- Ellis, M.E. and the Department of Transport (UK) 2019: Speed limit change helps increase productivity. UK Government: <https://www.gov.uk/government/news/speed-limit-change-helps-increase-productivity>
- Gossling, s., Kees, J., Litman, T., Humpe, A. 2023: The economic cost of a 130 kph speed limit in Germany. Ecological Economics 209: 107850. <https://www.sciencedirect.com/science/article/pii/S0921800923001131>
- Kingham, S. 2022: Lower speed limits don't just save lives, they make NZ towns and cities better places to live. The Conversation, November 17, <https://theconversation.com/lower-speed-limits-dont-just-save-lives-they-make-nz-towns-and-cities-better-places-to-live-194448>
- Kingham, S. 2024: 5 reasons why the Government should slow down on raising speed limits. The Conversation, July 2, <https://theconversation.com/give-way-5-reasons-why-the-government-should-slow-down-on-raising-speed-limits-233774>
- Quddus. M., Theofilatos, A., Feng, M. and Elvik, R. 2025: Evaluating the safety and speed impacts of the 20 mph speed limit in the UK: evidence and insights. Accident Analysis and Prevention 221: 108210. <https://doi.org/10.1016/j.aap.2025.108210>
- Waka Kotahi, 2024: <https://www.nzta.govt.nz/safety/driving-safely/speed/state-highway-speed-management/speed-reversals-and-consultation-transitional-changes-in-2024-25>
- Welch, T. 2025: Evidence shows that higher speed limits don't make sense. University of Auckland News and Opinion: [Evidence shows higher speed limits don't make financial sense – University of Auckland](https://www.auckland.ac.nz/en/news-and-opinion/evidence-shows-higher-speed-limits-dont-make-financial-sense-university-of-auckland)
- Yannis, G and Michelaraki, E. 2024: Review of city-wide 30 km/h speed limit benefits in Europe. Sustainability, 16: 4382. <https://www.mdpi.com/2071-1050/16/11/4382>
- Yannis, G and Michelaraki, E. 2025: Effectiveness of 30 km/h speed limit – a literature review. Journal of Safety research 92: 490-503. <https://www.sciencedirect.com/science/article/abs/pii/S002243752400152X?via%3Dihub>